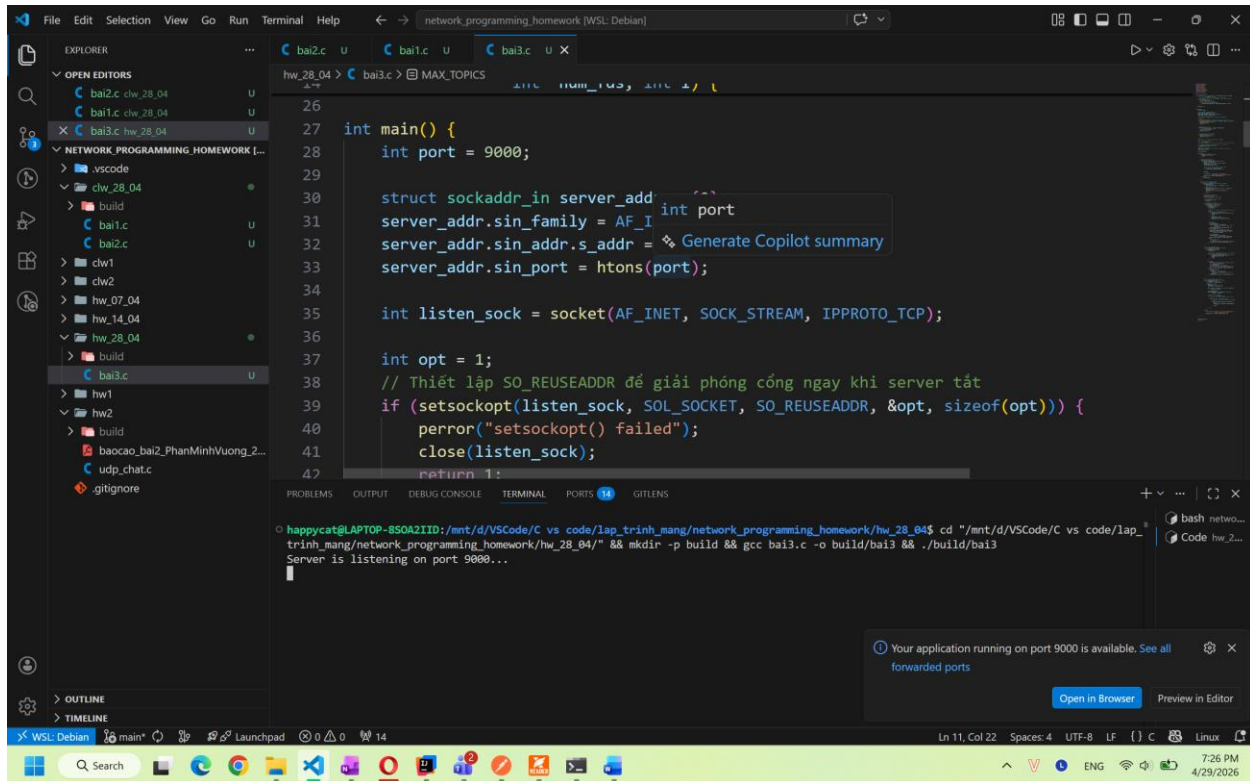


BÁO CÁO BÀI TẬP VỀ NHÀ NGÀY 28/04/2026 (Server định tuyến)

HỌ VÀ TÊN: PHAN MINH VƯỢNG

MSSV: 20225241

- Chạy server ở cổng 9000



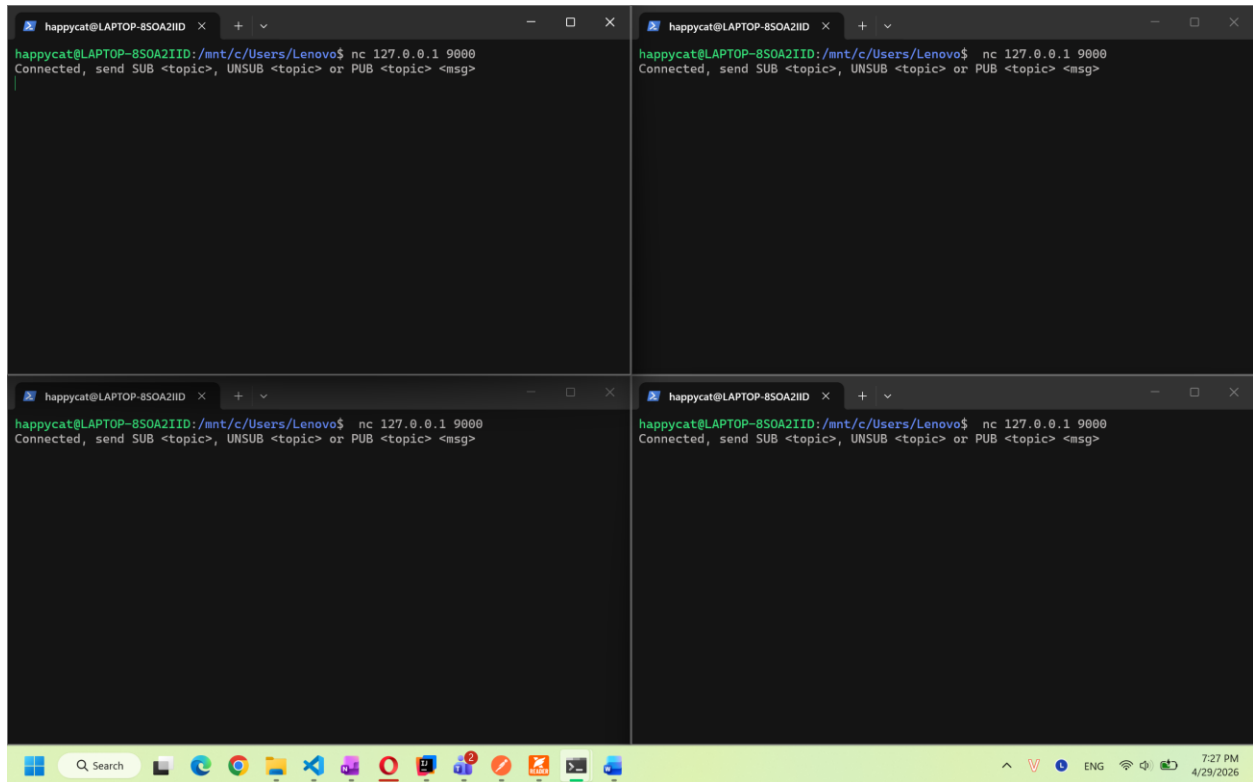
```
26
27 int main() {
28     int port = 9000;
29
30     struct sockaddr_in server_addr;
31     server_addr.sin_family = AF_INET;
32     server_addr.sin_addr.s_addr = INADDR_ANY;
33     server_addr.sin_port = htons(port);
34
35     int listen_sock = socket(AF_INET, SOCK_STREAM, IPPROTO_TCP);
36
37     int opt = 1;
38     // Thiết lập SO_REUSEADDR để giải phóng cổng ngay khi server tắt
39     if (setsockopt(listen_sock, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt))) {
40         perror("setsockopt() failed");
41         close(listen_sock);
42         return 1;
43     }
44 }
```

happy@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_28_04\$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_28_04/" && mkdir -p build && gcc bai3.c -o build/bai3 && ./build/bai3

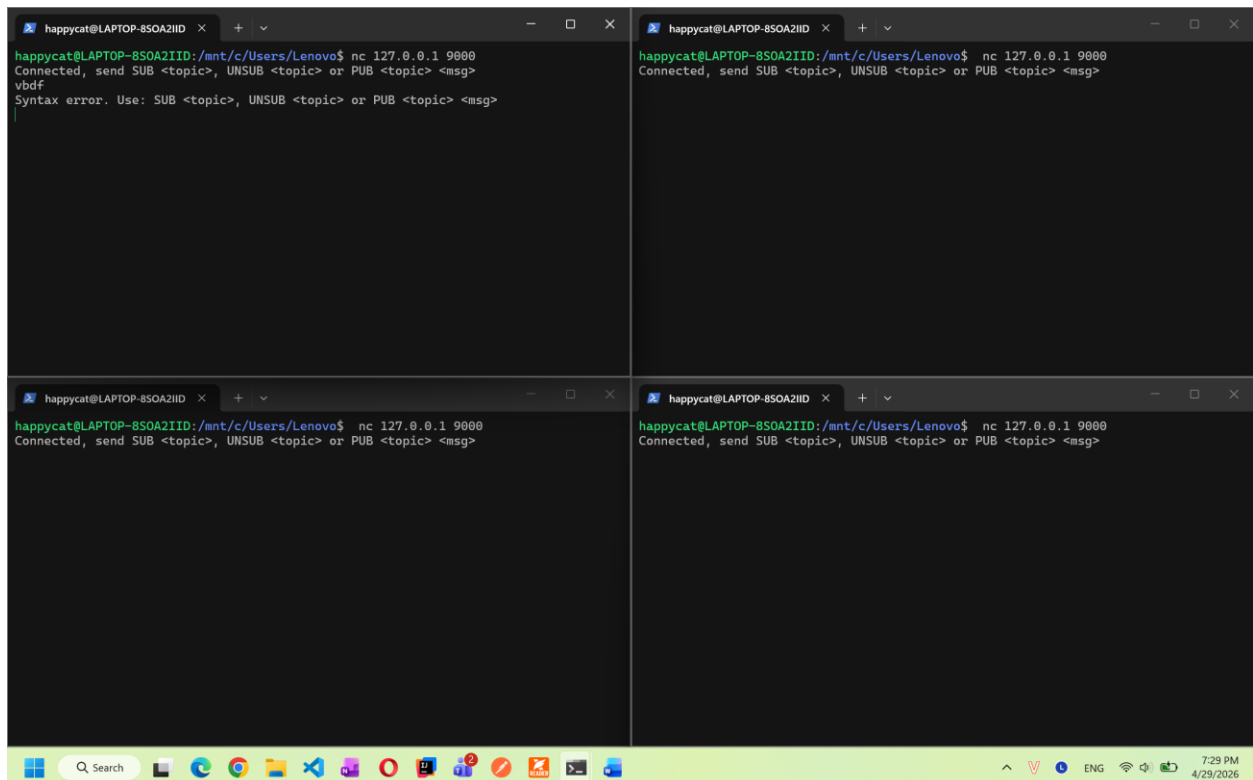
Server is listening on port 9000...

Your application running on port 9000 is available. See all forwarded ports.

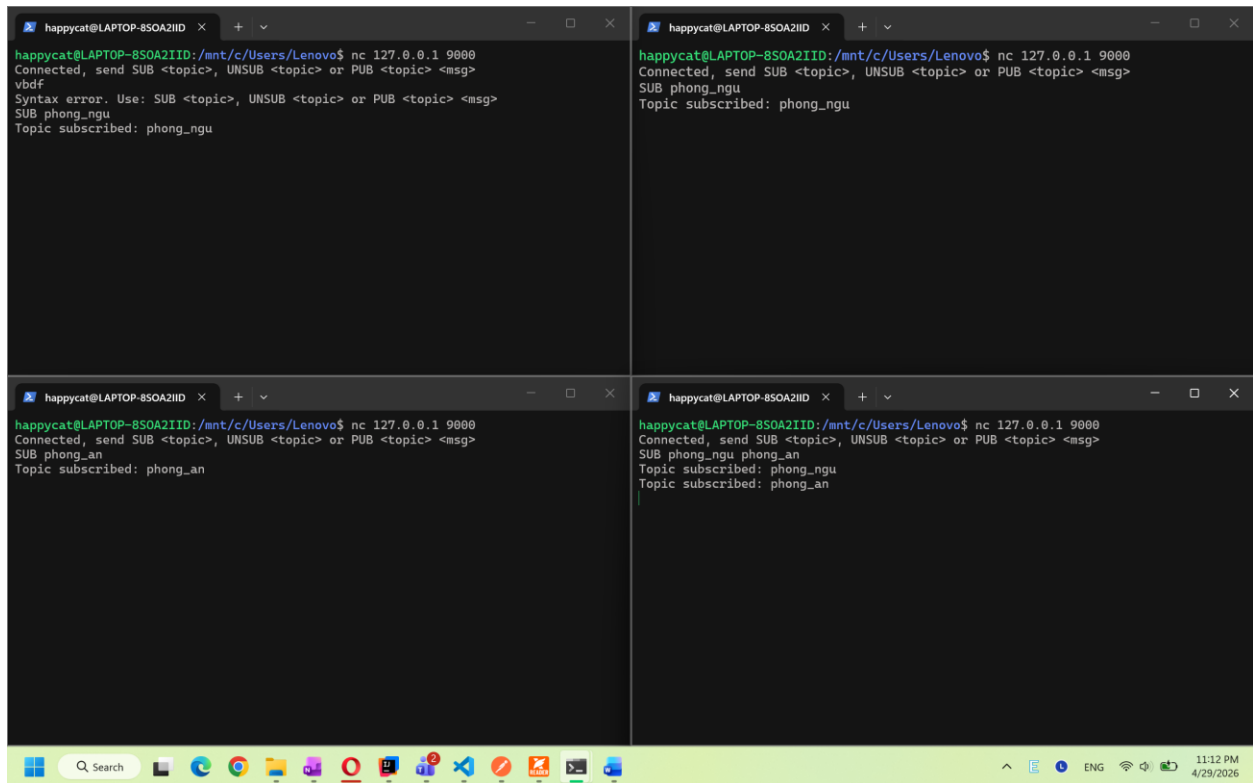
- Khởi tạo 4 client kết nối vào server



- Nếu client nhập sai, server sẽ gửi thông báo



- Cho client 1 và 2 subscribe topic “phong_ngu”, client 3 subscribe topic “phong_an”, client 4 subscribe 2 topic là "phong_ngu" và “phong_an”

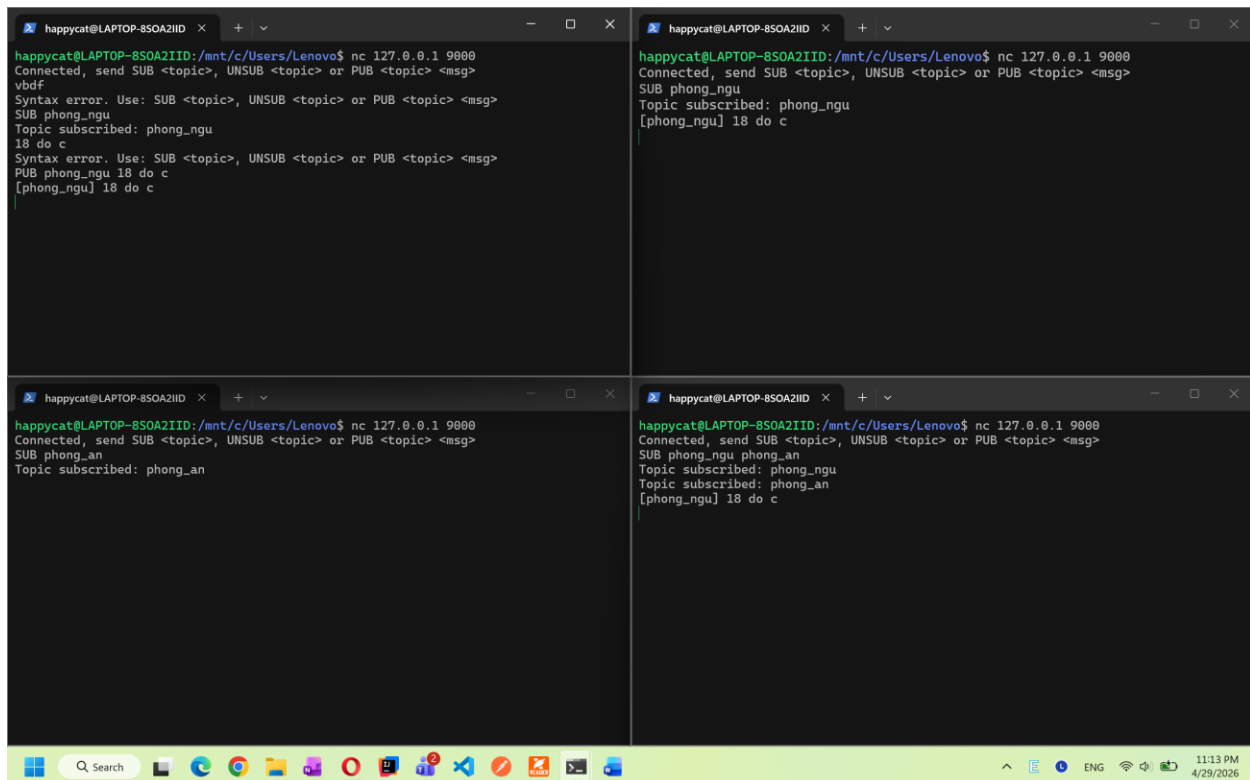


The image shows four terminal windows arranged in a 2x2 grid, each running the 'happy' MQTT broker. The windows demonstrate the setup of four clients:

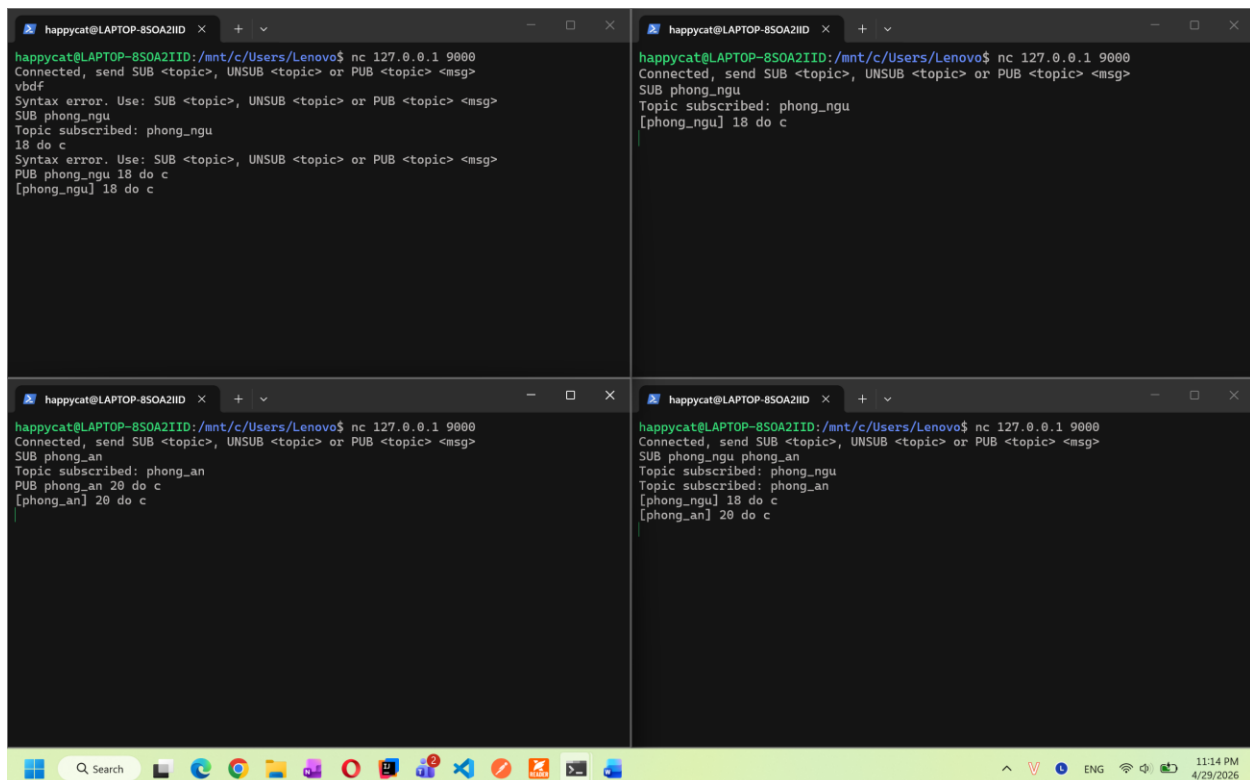
- Top-left window:** Client 1 connects and attempts to subscribe to 'phong_ngu'. It receives a 'Syntax error' message because it used the wrong command format. The correct command 'SUB phong_ngu' is then entered, and the client successfully subscribes to the 'phong_ngu' topic.
- Top-right window:** Client 2 connects and successfully subscribes to the 'phong_ngu' topic using the correct 'SUB' command.
- Bottom-left window:** Client 3 connects and successfully subscribes to the 'phong_an' topic using the correct 'SUB' command.
- Bottom-right window:** Client 4 connects and successfully subscribes to both the 'phong_ngu' and 'phong_an' topics using the correct 'SUB' command.

The Windows taskbar at the bottom indicates the system time is 11:12 PM on 4/29/2026.

- Client 1 publish “18 do c” vào topic “phong_ngu” thì client 1,2,4 sẽ nhận được data



- Client 3 publish “20 do c” vào topic “phong_an” thì client 3,4 sẽ nhận được



- Client 4 unsub topic “phong_ngu”, sau đó client 1 publish “19 do c” vào topic “phong_ngu” thì chỉ có client 1,2 nhận được, client 4 không nhận được nữa

```
happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 9000
Connected, send SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
vddf
Syntax error. Use: SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
SUB phong_ngu
Topic subscribed: phong_ngu
18 do c
Syntax error. Use: SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
PUB phong_ngu 18 do c
[phong_ngu] 18 do c
PUB phong_ngu 19 do c
[phong_ngu] 19 do c

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 9000
Connected, send SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
SUB phong_ngu
Topic subscribed: phong_ngu
[phong_ngu] 18 do c
[phong_ngu] 19 do c

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 9000
Connected, send SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
SUB phong_an
Topic subscribed: phong_an
PUB phong_an 20 do c
[phong_an] 20 do c

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 9000
Connected, send SUB <topic>, UNSUB <topic> or PUB <topic> <msg>
SUB phong_ngu phong_an
Topic subscribed: phong_ngu
Topic subscribed: phong_an
[phong_ngu] 18 do c
[phong_an] 20 do c
UNSUB phong_ngu
Topic unsubscribed: phong_ngu
```

- Nếu client subscribe vào topic đã sub trước đó thì sẽ nhận được thông báo

The image displays four terminal windows, each running a C program named 'happycat' on a Windows laptop (LAPTOP-850A2IID). The programs are connected to an MQTT broker at 127.0.0.1:9000. The interactions are as follows:

- Top-left:** The client subscribes to the 'phong_ngu' topic. It receives two messages: '[phong_ngu] 18 do c' and '[phong_ngu] 19 do c'. It then publishes two messages: 'PUB phong_ngu 18 do c' and 'PUB phong_ngu 19 do c'.
- Top-right:** The client subscribes to the 'phong_ngu' topic. It receives two messages: '[phong_ngu] 18 do c' and '[phong_ngu] 19 do c'.
- Bottom-left:** The client subscribes to the 'phong_an' topic. It receives two messages: '[phong_an] 20 do c' and '[phong_an] 20 do c'.
- Bottom-right:** The client subscribes to both 'phong_ngu' and 'phong_an' topics. It receives messages for both: '[phong_ngu] 18 do c', '[phong_an] 20 do c', and '[phong_ngu] 20 do c'. It then unsubscribes from the 'phong_ngu' topic.

- Kết quả phía server

The image shows a VS Code editor window with a C program for an MQTT server. The code is located in 'network_programming_homework/hw_28_04/bai3.c'. The terminal output shows the server's activity:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_28_04$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_28_04/" && mkdir -p build && gcc bai3.c -o build/bai3 && ./build/bai3
Server is listening on port 9000...
New client connected: 4
New client connected: 5
New client connected: 6
New client connected: 7
Client 4 subscribed to topic: 'phong_ngu'
Client 5 subscribed to topic: 'phong_ngu'
Client 6 subscribed to topic: 'phong_an'
Client 7 subscribed to topic: 'phong_ngu'
Client 7 subscribed to topic: 'phong_an'
Client 4 published to 'phong_ngu': 18 do c
Client 6 published to 'phong_an': 20 do c
Client 7 unsubscribed from topic: 'phong_ngu'
Client 4 published to 'phong_ngu': 19 do c
```